



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

Instructor: Prashant Kumar Pandey



Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks



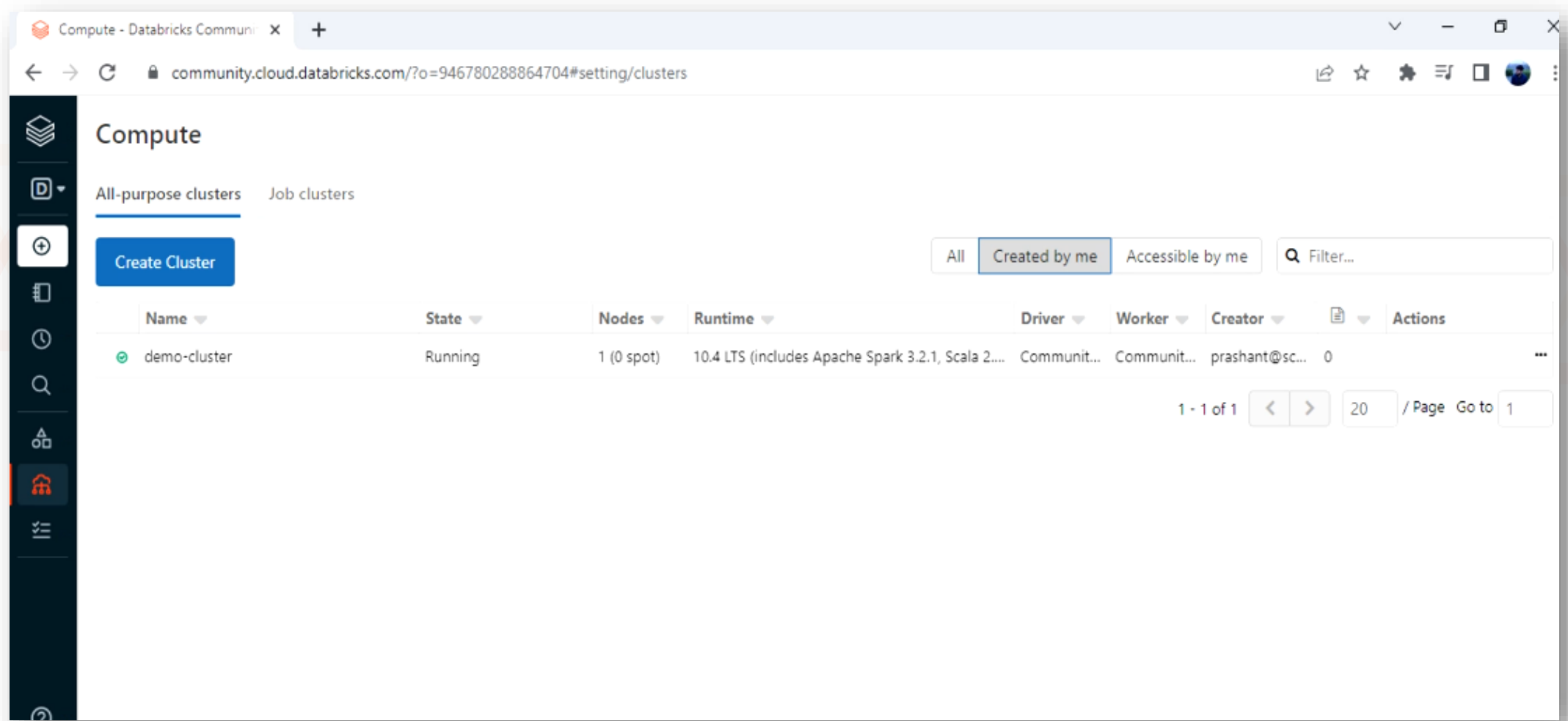
Spark Web UI - Introduction



Spark Web UI

The go-to place to monitor the status and resource consumption of your Spark cluster

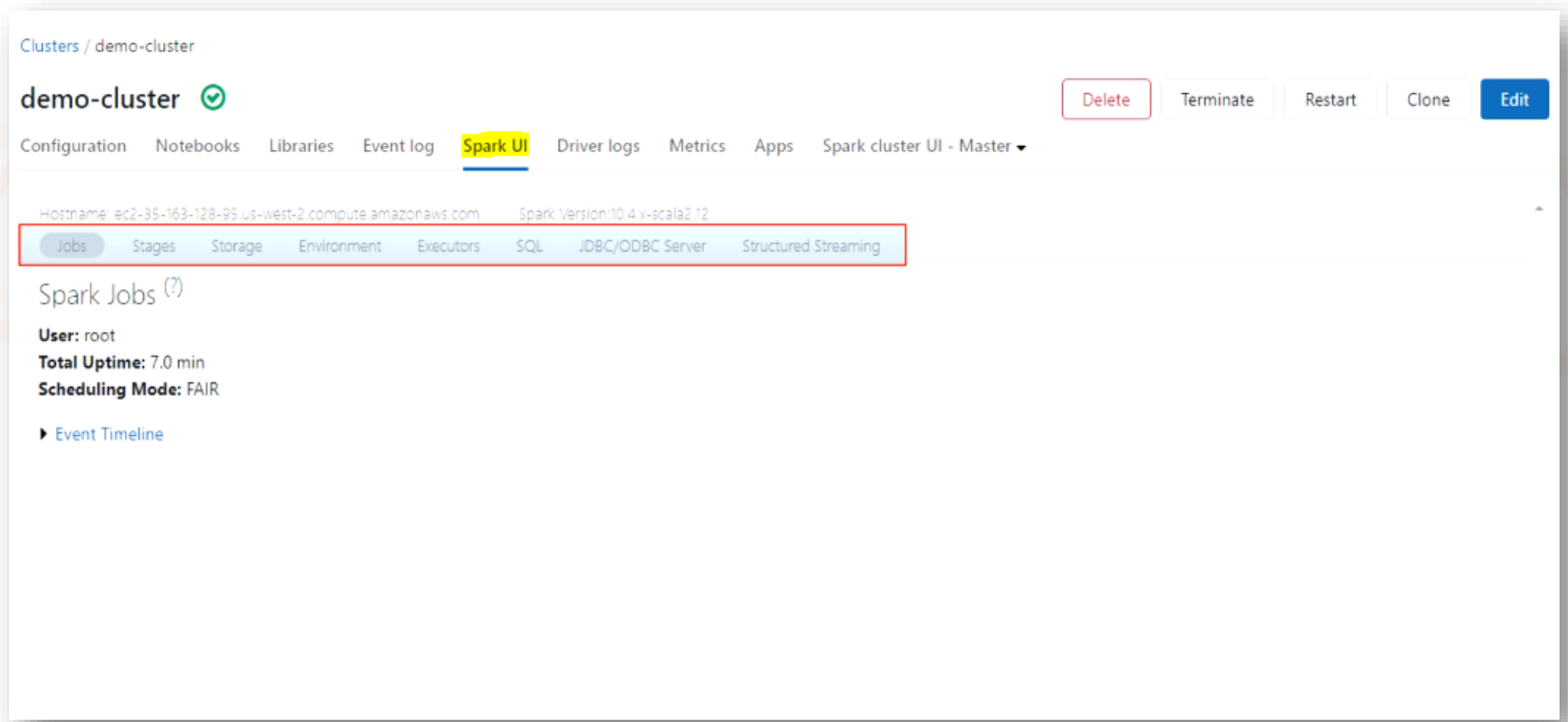
Go to your Databricks workspace.
Now, go to compute menu and start a cluster.
I already have one cluster running.
Click the cluster and go to the cluster details page.



Then, navigate to your Spark Web UI. You will see some tabs such as Jobs, Stages, storage, etc. Spark UI is your go-to place for monitoring your Spark applications and understanding many other details about your application.

You do not see anything here because I haven't run anything on this cluster.

Things will start appearing here as you start running your application or the Spark code.



Open a notebook from the previous lecture. **(Reference - 09-understanding-table-types)**
Attach a cluster to this notebook. And run all the cells up to where we are writing a
Dataframe.

```
09-understanding-table-types Python

1 flight_schema_ddl = """FL_DATE STRING, OP_CARRIER STRING, OP_CARRIER_FL_NUM INT, ORIGIN STRING,
2     ORIGIN_CITY_NAME STRING, DEST STRING, DEST_CITY_NAME STRING, CRS_DEP_TIME INT, DEP_TIME INT,
3     WHEELS_ON INT, TAXI_IN INT, CRS_ARR_TIME INT, ARR_TIME INT, CANCELLED STRING, DISTANCE INT"""

Command took 0.06 seconds -- by prashant@scholarnest.com at 5/3/2022, 8:35:44 PM on demo-cluster

1 flight_time_raw_df = spark.read \
2     .format("json") \
3     .schema(flight_schema_ddl) \
4     .option("mode", "FAILFAST") \
5     .option("dateFormat", "M/d/y") \
6     .load("/FileStore/tables/flight_time.json")

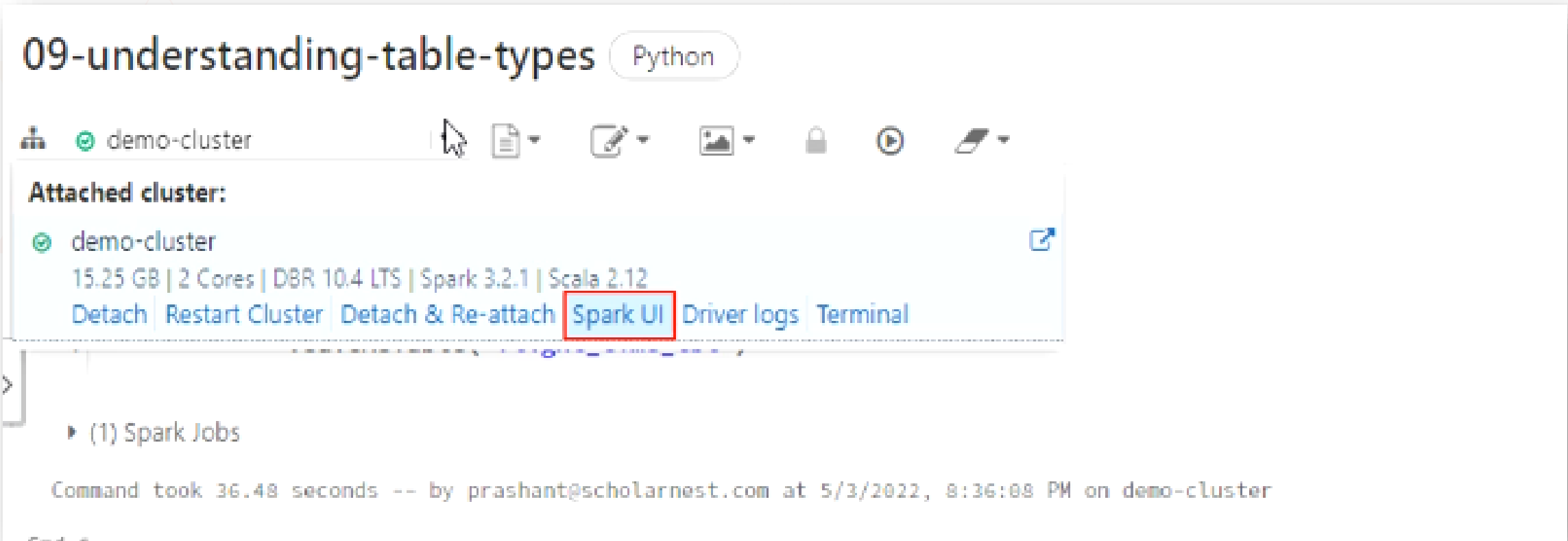
Command took 2.19 seconds -- by prashant@scholarnest.com at 5/3/2022, 8:35:46 PM on demo-cluster

1 from pyspark.sql.functions import to_date, expr
2
3 flight_time_df = flight_time_raw_df \
4     .withColumn("FL_DATE", to_date("FL_DATE", "M/d/y")) \
5     .withColumn("CANCELLED", expr("if(CANCELLED=1, true, false)))

Command took 0.44 seconds -- by prashant@scholarnest.com at 5/3/2022, 8:35:50 PM on demo-cluster

Cmd 4
```

Click the demo-cluster drop down, and you will see a direct link for the Spark UI.
Let me go there.



You can see some information here. We are currently in the Jobs tab of the Spark UI. The Jobs tab displays a summary page of all jobs in the Spark application. The summary page shows high-level information, such as the status, duration, and progress bar.

09-understanding-table-types - X

Databricks Shell - Spark Jobs

X

+

community.cloud.databricks.com/sparkui/0503-145506-vzyghp07/driver-6906967366313043515/?csrf=32012ed9-cdce-43ac-bc01-fc9dd70eee3a&o=946780...

JobsStagesStorageEnvironmentExecutorsSQLJDBC/ODBC ServerStructured Streaming

Spark Jobs (?)

User: root

Total Uptime: 11 min

Scheduling Mode: FAIR

Completed Jobs: 1

Event Timeline

Completed Jobs (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Job Id (Job Group) ▾	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
0 (1901202411884247856_5170693221701987476_a5f884691dec46bd892089b3a869635e)	flight_time_df.write \.format("par... saveAsTable at NativeMethodAccessorImpl.java:0	2022/05/03 15:06:10	30 s	1/1	8/8

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

You can click on a job description on the summary page to see the details.

JobsStagesStorageEnvironmentExecutorsSQLJDBC/ODBC ServerStructured Streaming

Spark Jobs (?)

User: root

Total Uptime: 11 min

Scheduling Mode: FAIR

Completed Jobs: 1

▶ Event Timeline

▼ Completed Jobs (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Job Id (Job Group) ▼	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
0 (1901202411884247856_5170693221701987476_a5f884691dec46bd892089b3a869635e)	flight_time_df.write \.format("par... saveAsTable at NativeMethodAccessorImpl.java:0	2022/05/03 15:06:10	30 s	1/1	8/8

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

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Here we are on the Job details for job id zero.

The Job details page shows you the stage-wise breakup of your Job. I have only one stage in this Job. The details page also shows some other information about the Job stages, such as duration and task progress bar.

Jobs

Stages

Storage

Environment

Executors

SQL

JDBC/ODBC Server

Structured Streaming

Details for Job 0

Status: SUCCEEDED

Submitted: 2022/05/03 15:06:10

Duration: 30 s

Associated SQL Query: 4

Job Group: 1901202411884247856_5170693221701987476_a5f884691dec46bd892089b3a869635e

Completed Stages: 1

▶ Event Timeline

▶ DAG Visualization

▼Completed Stages (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

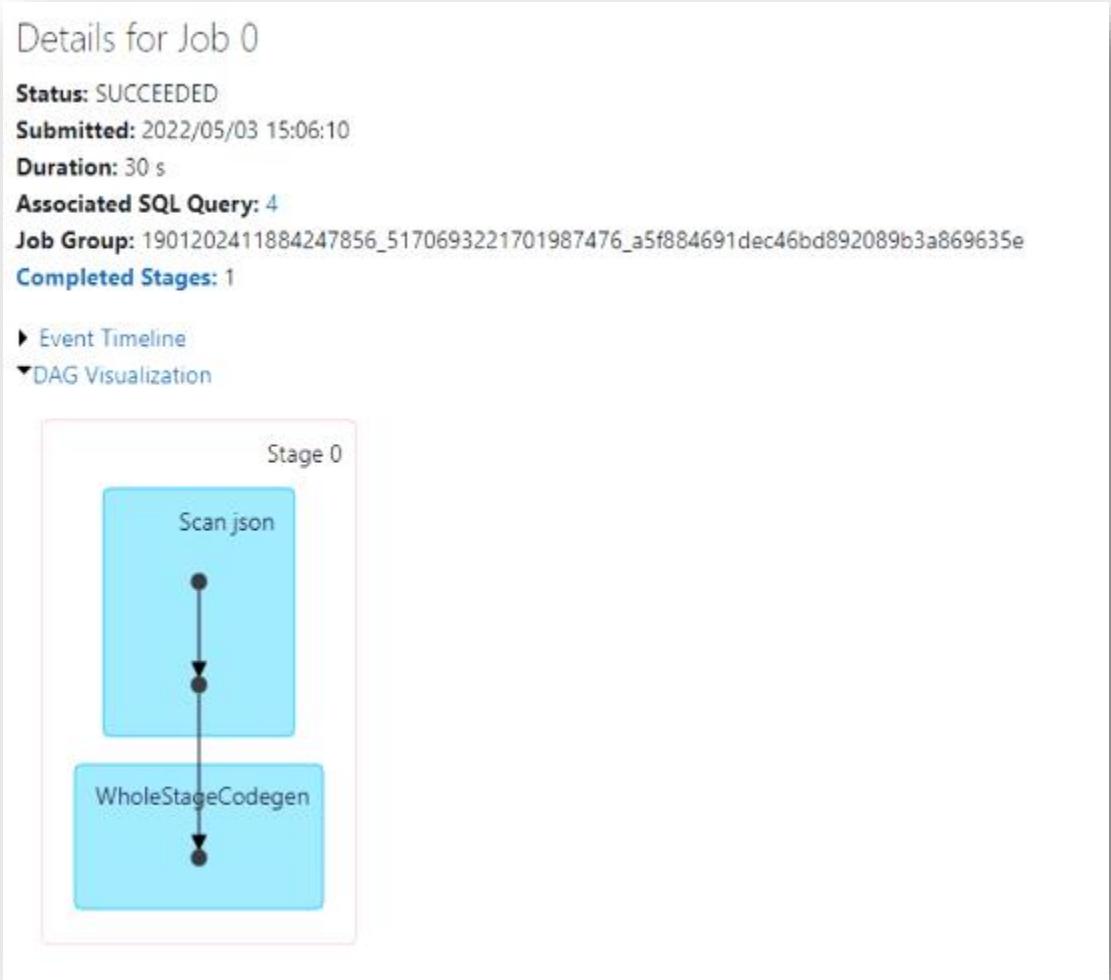
Stage Id ▼	Pool Name	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
0	1901202411884247856	flight_time_df.write \.format("par... saveAsTable at NativeMethodAccessorImpl.java:0 +details	2022/05/03 15:06:11	30 s	8/8		3.6 MiB		

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Click the DAG visualization link to see the DAG for the selected Job.

You might not understand these things yet because I didn't cover Spark Jobs, Stages, and Tasks. However, the information shown over the Spark UI is crucial for understanding your application behavior and resources and identifying tuning and optimization opportunities.



I have only one Job as of now. It makes sense to run a few more jobs to understand how this page looked when we ran multiple jobs on the same cluster. So come back to your notebook and run a few more commands. Run the create external table command, come down to the select count(*) from ext_flight_time_tbl, and run that as well.

Now, look at the output. It says 2 Spark Jobs and also shows the count. So the select statement triggered two Spark jobs.

Cmd 14

1 %sql

2 select count(*) from ext_flight_time_tbl

▶ (2) Spark Jobs

	count(1)	
1	300000	

Showing all 1 rows.

Command took 9.85 seconds -- by prashant@scholarnest.com at 5/3/2022, 8:42:44 PM on demo-cluster

Go back to the Spark UI once again and refresh the page, and you will see three jobs. We can see Job IDs in the first column. Job-0 was triggered by the saveAsTable() method. The saveAsTable() is a DataFramewriter method. Job-1 and Job-2 were triggered by the select count(*) SQL.

09-understanding-table-types - x Databricks Shell - Spark Jobs +

community.cloud.databricks.com/sparkui/0503-145506-vzyghp07/driver-6906967366313043515/jobs/?o=946780288864704

Jobs Stages Storage Environment Executors SQL JDBC/ODBC Server Structured Streaming

Spark Jobs (?)

User: root

Total Uptime: 18 min

Scheduling Mode: FAIR

Completed Jobs: 3

Event Timeline

Completed Jobs (3)

Page: 1

1 Pages. Jump to 1 . Show 100 items in a page. Go

Job Id (Job Group) ▼	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2 (3094848351750946294_5576904425099990681_042e26e8d430464e88e38248396474c1)	select count(*) from ext_flight_time_tbl collectResult at OutputAggregator.scala:268	2022/05/03 15:12:53	0.3 s	1/1 (1 skipped)	1/1 (8 skipped)
1 (3094848351750946294_5576904425099990681_042e26e8d430464e88e38248396474c1)	select count(*) from ext_flight_time_tbl collectResult at OutputAggregator.scala:268	2022/05/03 15:12:45	7 s	1/1	8/8
0 (1901202411884247856_5170693221701987476_a5f884691dec46bd892089b3a869635e)	flight_time_df.write \ .format("par... saveAsTable at NativeMethodAccessorImpl.java:0	2022/05/03 15:06:10	30 s	1/1	8/8

Page: 1

1 Pages. Jump to 1 . Show 100 items in a page. Go

Now, go to the next tab that shows the stage summary page.

So the stage summary page shows all the stages across the Jobs.

We ran three jobs, and we saw them on the Jobs summary page.

Altogether, those four jobs ran in four stages, and we can see four stages listed here on this stages summary page.

Jobs **Stages** Storage Environment Executors SQL JDBC/ODBC Server Structured Streaming

Completed Stages: 3
Skipped Stages: 1

▼Fair Scheduler Pools (1)

Pool Name	Minimum Share	Pool Weight	Active Stages	Running Tasks	SchedulingMode
default	0	1	0	0	FIFO

▼Completed Stages (3)

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Pool Name	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
3	3094848351750946294	select count(*) from ext_flight_time_tbl collectResult at OutputAggregator.scala:268 +details	2022/05/03 15:12:53	0.2 s	1/1			472.0 B	
1	3094848351750946294	select count(*) from ext_flight_time_tbl collectResult at OutputAggregator.scala:268 +details	2022/05/03 15:12:46	7 s	8/8				472.0 B
0	1901202411884247856	flight_time_df.write \.format("par... saveAsTable at NativeMethodAccessorImpl.java:0 +details	2022/05/03 15:06:11	30 s	8/8		3.6 MiB		

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

▼Skipped Stages (1)

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Pool Name	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
2	default	collectResult at OutputAggregator.scala:268 +details	Unknown	Unknown	0/8				

Go to the Jobs tab.

I have three jobs, and I know Job-0 is saving my Dataframe to a Spark table.

I also know that Job-1 and Job-2 give me counts from the external table.

The Job-0 took 30 seconds, so I want to investigate it further.

So I will click the Job description and reach the Job details page.

Jobs

Stages

Storage

Environment

Executors

SQL

JDBC/ODBC Server

Structured Streaming

Spark Jobs (?)

User: root

Total Uptime: 43 min

Scheduling Mode: FAIR

Completed Jobs: 3

▶ Event Timeline

▼ Completed Jobs (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Job Id (Job Group) ▼	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2 (3094848351750946294_5576904425099990681_042e26e8d430464e88e38248396474c1)	select count(*) from ext_flight_time_tbl collectResult at OutputAggregator.scala:268	2022/05/03 15:12:53	0.3 s	1/1 (1 skipped)	1/1 (8 skipped)
1 (3094848351750946294_5576904425099990681_042e26e8d430464e88e38248396474c1)	select count(*) from ext_flight_time_tbl collectResult at OutputAggregator.scala:268	2022/05/03 15:12:45	7 s	1/1	8/8
0 (1901202411884247856_5170693221701987476_a5f884691dec46bd892089b3a869635e)	flight_time_df.write \.format("par... saveAsTable at NativeMethodAccessorImpl.java:0	2022/05/03 15:06:10	30 s	1/1	8/8

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

I see only one stage here, and I know this stage belongs to Job-0.
I want to look into the stage details, so I will click the Stage description and go to the stage details page.

Jobs

Stages

Storage

Environment

Executors

SQL

JDBC/ODBC Server

Structured Streaming

Completed Stages: 1

▶ Event Timeline

▼ DAG Visualization

Stage 0

Scan json

WholeStageCodegen

▼ Completed Stages (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id ▼	Pool Name	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
0	1901202411884247856	flight_time_df.write \.format("par... saveAsTable at NativeMethodAccessorImpl.java:0 +details	2022/05/03 15:06:11	30 s	8/8		3.6 MiB		

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Here I am in the stage details page. I can see the stage details of my Job-0.
You can come here from the stage summary page also.

Jobs

Stages

Storage

Environment

Executors

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JDBC/ODBC Server

Structured Streaming

Details for Stage 0 (Attempt 0)

Resource Profile Id: 0

Total Time Across All Tasks: 3.8 min

Locality Level Summary: Process local: 8

Output Size / Records: 3.6 MiB / 300000

Associated Job Ids: 0

▶ DAG Visualization

▶ Show Additional Metrics

▶ Event Timeline

Summary Metrics for 8 Completed Tasks

Metric	Min	25th percentile	Median	75th percentile	Max
Duration	28 s	28 s	28 s	29 s	29 s
GC Time	2 s	2 s	2 s	2 s	2 s
Output Size / Records	316.1 KiB / 25090	473.7 KiB / 39257	480.7 KiB / 39276	486.7 KiB / 39351	489.3 KiB / 39379

Showing 1 to 3 of 3 entries

▶ Aggregated Metrics by Executor

Tasks (8)

Show 20 entries

Search:

Task Index	Task ID	Attempt	Status	Locality level	Executor ID	Host	Logs	Launch Time	Duration	GC Time	Output Size / Records	Errors
0	0	0	SUCCESS	PROCESS_LOCAL	driver	ip-10-172-228-131-us-west		2022-05-04	28 s	2 s	489.3 KiB / 39379	

We also have a storage tab.

We do not see enough information here, but the storage page is your gateway to see the cache information.

This page shows information when you cache something in Spark memory

JobsStagesStorageEnvironmentExecutorsSQLJDBC/ODBC ServerStructured Streaming

Storage

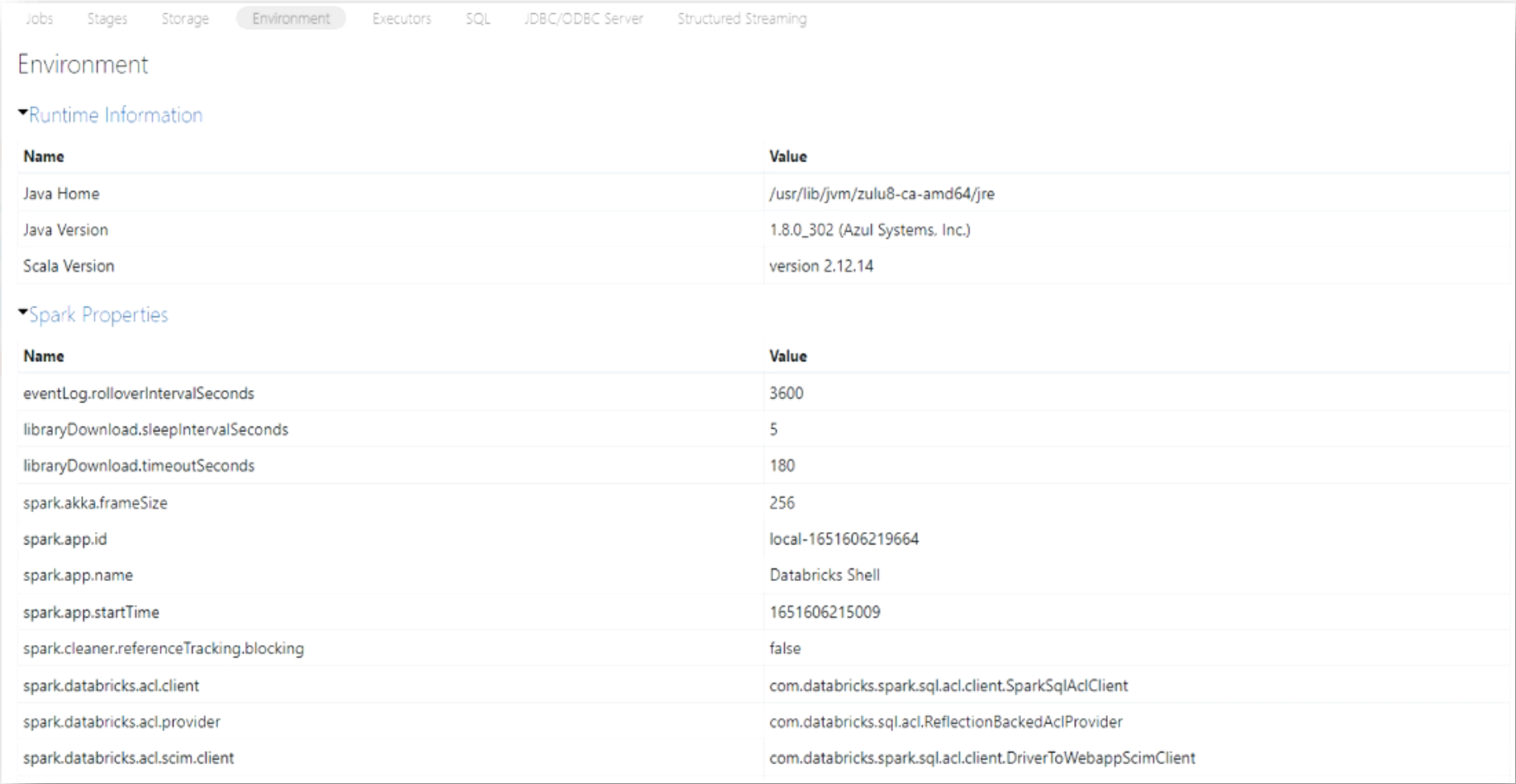
Parquet IO Cache

Data Read from External Filesystem (All Formats)	Data Read from IO Cache (Cache Hits, Compressed)	Data Written to IO Cache (Compressed)	Cache Misses (Compressed)	True Cache Misses	Partial Cache Misses	Rescheduling Cache Misses	Cache Hit Ratio	Number of Local Scan Tasks	Number of Rescheduled Scan Tasks	Cache Metadata Manager Peak Disk Usage
83.6 MiB	0.0 B	0.0 B	0.0 B	0.0 B	0.0 B	0.0 B	0 %	0	0	0.0 B

The next page is for the environment.

The Environment tab displays the values for the different environment and configuration variables, including JVM, Spark, and system properties.

We refer to this page to understand the runtime environment and see if some tuning is required.



The screenshot shows the 'Environment' tab in the Databricks interface. It contains two sections: 'Runtime Information' and 'Spark Properties', each with a table of configuration variables.

Environment	
▼ Runtime Information	
Name	Value
Java Home	/usr/lib/jvm/zulu8-ca-amd64/jre
Java Version	1.8.0_302 (Azul Systems, Inc.)
Scala Version	version 2.12.14
▼ Spark Properties	
Name	Value
eventLog.rolloverIntervalSeconds	3600
libraryDownload.sleepIntervalSeconds	5
libraryDownload.timeoutSeconds	180
spark.akka.frameSize	256
spark.app.id	local-1651606219664
spark.app.name	Databricks Shell
spark.app.startTime	1651606215009
spark.cleaner.referenceTracking.blocking	false
spark.databricks.acl.client	com.databricks.spark.sql.acl.client.SparkSqlAclClient
spark.databricks.acl.provider	com.databricks.sql.acl.ReflectionBackedAclProvider
spark.databricks.acl.scim.client	com.databricks.spark.sql.acl.client.DriverToWebappScimClient

The next tab is the executor tab.

This guy displays summary information about the executors, including memory, cores, disk usage, and shuffle information each executor uses.

Jobs

Stages

Storage

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JDBC/ODBC Server

Structured Streaming

Executors

Show Additional Metrics

Summary

	▲ RDD Blocks ↕	Storage Memory ↕	Disk Used ↕	Cores ↕	Active Tasks ↕	Failed Tasks ↕	Complete Tasks ↕	Total Tasks ↕	Task Time (GC Time) ↕	Input ↕	Shuffle Read ↕	Shuffle Write ↕	Excluded ↕
Active(1)	0	260.5 KiB / 3.9 GiB	0.0 B	8	0	0	17	17	5.1 min (17 s)	0.0 B	472 B	472 B	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	260.5 KiB / 3.9 GiB	0.0 B	8	0	0	17	17	5.1 min (17 s)	0.0 B	472 B	472 B	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	▲ RDD Blocks ↕	Storage Memory ↕	Disk Used ↕	Cores ↕	Active Tasks ↕	Failed Tasks ↕	Complete Tasks ↕	Total Tasks ↕	Task Time (GC Time) ↕	Input ↕	Shuffle Read ↕	Shuffle Write ↕	Thread Dump ↕	Heap Histogram ↕	Exec Loss Reason ↕
driver	10.172.241.6:38291	Active	0	260.5 KiB / 3.9 GiB	0.0 B	8	0	0	17	17	5.1 min (17 s)	0.0 B	472 B	472 B	Thread Dump	Heap Histogram	

Showing 1 to 1 of 1 entries

Previous

Next

The next page is for SQL.

This page shows the list of SQL statements executed on your cluster.

You can see the time taken by each SQL. It is a summary page, and you go to the SQL details page, clicking any SQL.

JobsStagesStorageEnvironmentExecutorsSQLJDBC/ODBC ServerStructured Streaming

SQL

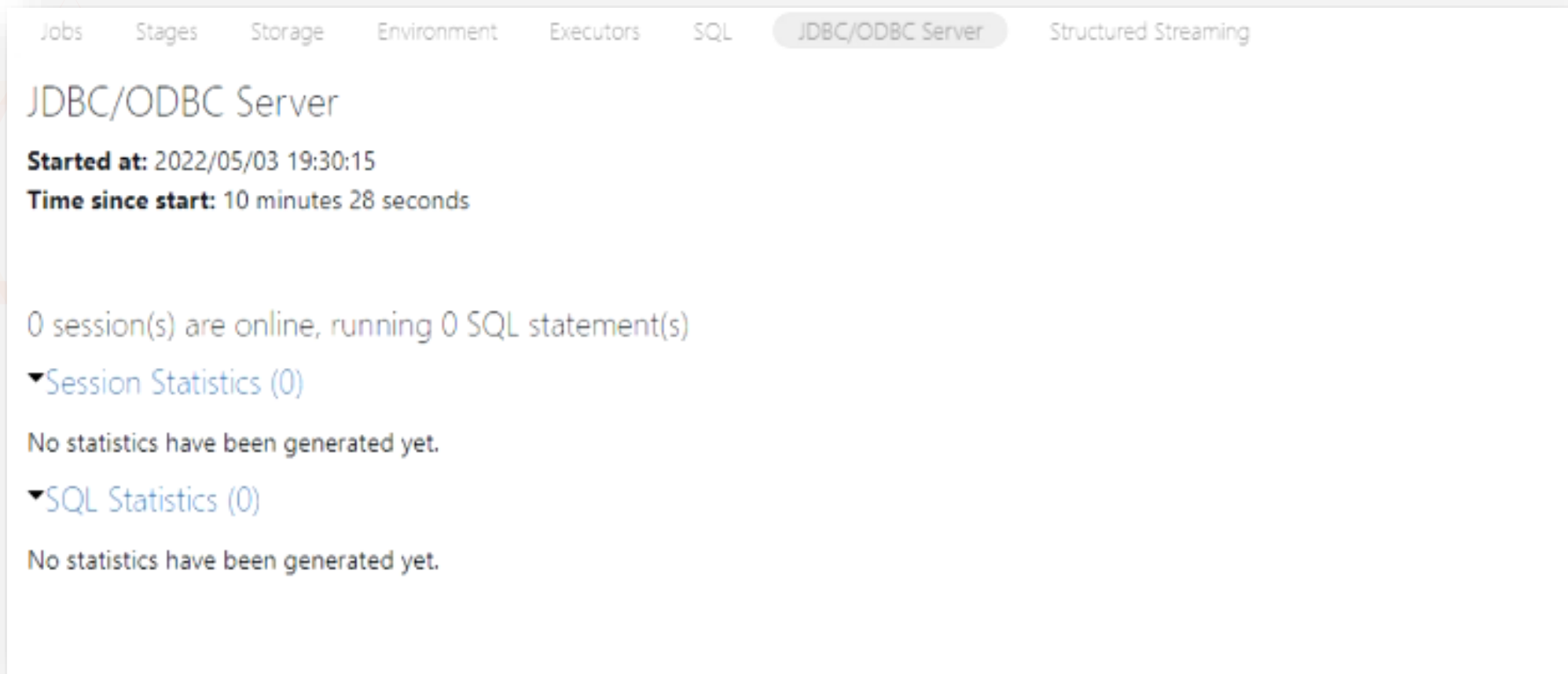
Completed Queries: 8

▼Completed Queries (8)

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

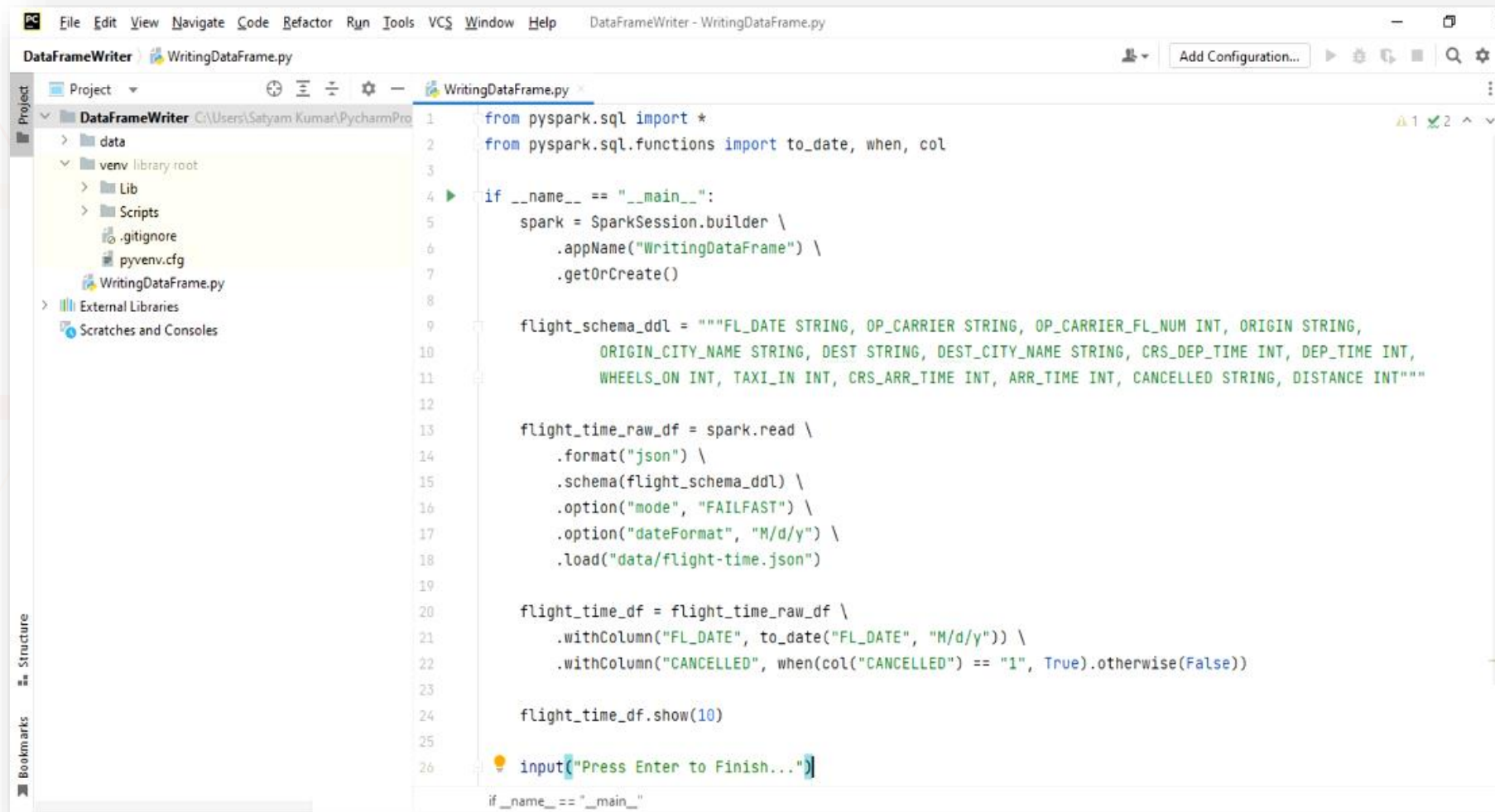
ID ▾	Description		Submitted	Duration	Job IDs
7	select count(*) from ext_flight_time_tbl	+details	2022/05/03 19:35:54	8 s	[1][2]
6	create table if not exists ext_flight_time_tbl ...	+details	2022/05/03 19:35:46	9 ms	
5	create table if not exists ext_flight_time_tbl ...	+details	2022/05/03 19:35:45	0.7 s	
4	flight_time_df.write \.format("par...	+details	2022/05/03 19:34:34	39 s	[0]
3	show tables in `default`	+details	2022/05/03 19:34:21	42 ms	
2	show tables in `default`	+details	2022/05/03 19:34:20	0.2 s	
1	show databases	+details	2022/05/03 19:34:15	0.1 s	
0	show databases	+details	2022/05/03 19:33:57	17 s	

You also have JDBC/ODBC server tab.
This tab is visible only when your Spark cluster also runs a Thrift server and allows JDBC/ODBC connections.
We do not see anything here because we do not have any connections.
The last tab is for Spark Structured Streaming.
This tab shows information about streaming queries and micro-batches.
We will cover this tab in the Spark streaming course.



You create and test the Spark application on your local machine. And you can still access the Spark UI. Here is my PyCharm IDE. I created a new project and wrote some code here. **(Reference - DataFramewriter)**

It is the same code that we saw in the notebook.

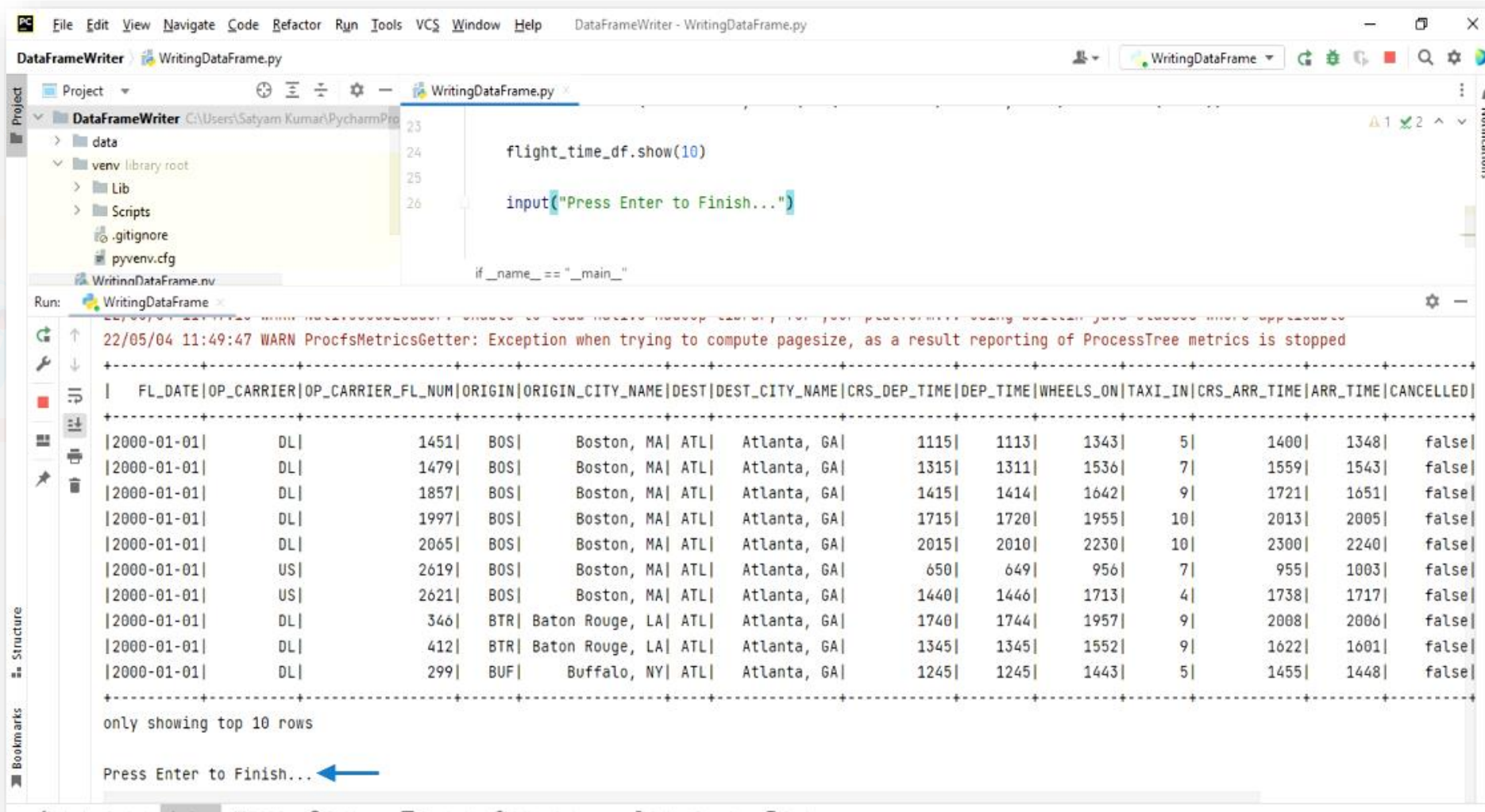


The screenshot shows the PyCharm IDE interface. The top menu bar includes File, Edit, View, Navigate, Code, Refactor, Run, Tools, VCS, Window, and Help. The title bar indicates the file is 'DataFrameWriter - WritingDataFrame.py'. The left sidebar shows the project structure for 'DataFrameWriter' located at 'C:\Users\Satyam Kumar\PycharmPro'. The structure includes a 'data' folder, a 'venv' folder (library root), and 'Lib' and 'Scripts' subfolders. The main editor window displays the following Python code:

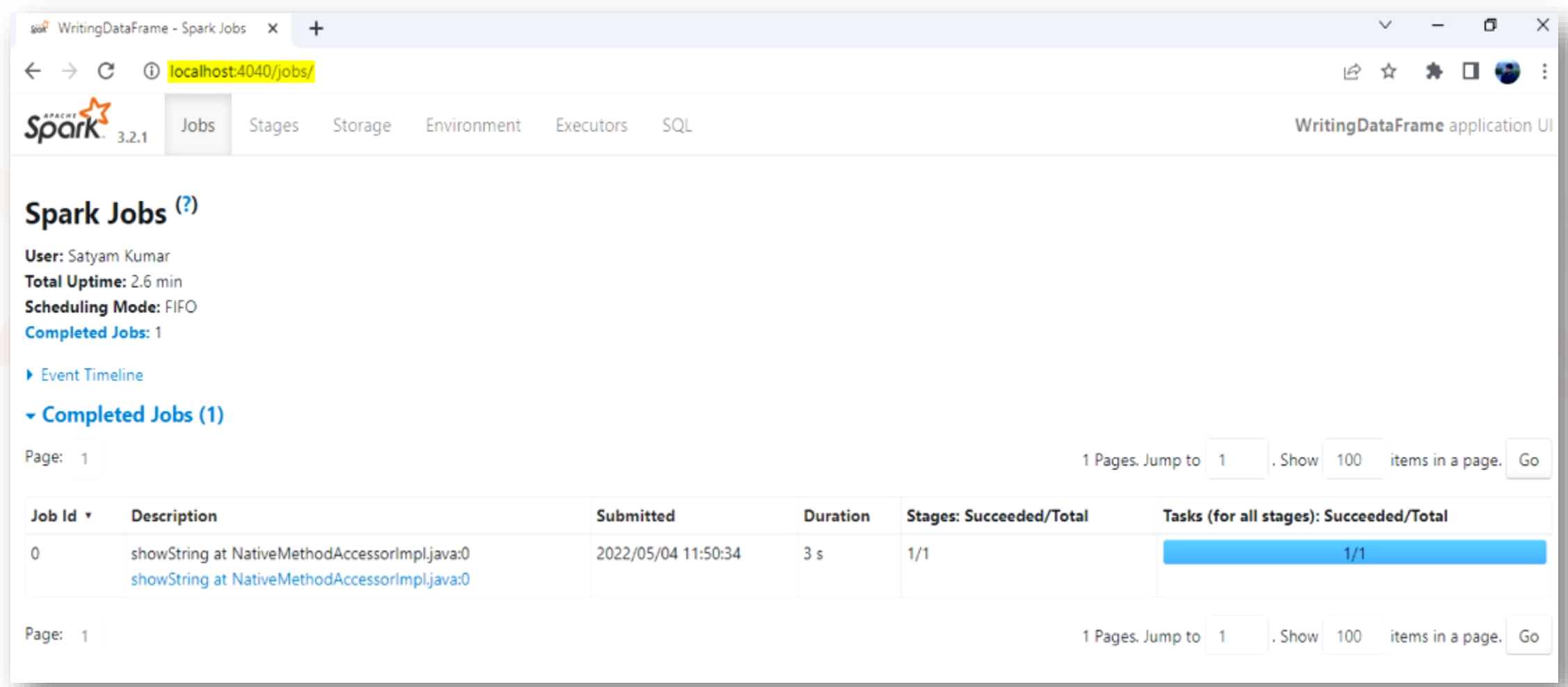
```
1 from pyspark.sql import *
2 from pyspark.sql.functions import to_date, when, col
3
4 if __name__ == "__main__":
5     spark = SparkSession.builder \
6         .appName("WritingDataFrame") \
7         .getOrCreate()
8
9     flight_schema_ddl = """FL_DATE STRING, OP_CARRIER STRING, OP_CARRIER_FL_NUM INT, ORIGIN STRING,
10        ORIGIN_CITY_NAME STRING, DEST STRING, DEST_CITY_NAME STRING, CRS_DEP_TIME INT, DEP_TIME INT,
11        WHEELS_ON INT, TAXI_IN INT, CRS_ARR_TIME INT, ARR_TIME INT, CANCELLED STRING, DISTANCE INT"""
12
13     flight_time_raw_df = spark.read \
14         .format("json") \
15         .schema(flight_schema_ddl) \
16         .option("mode", "FAILFAST") \
17         .option("dateFormat", "M/d/y") \
18         .load("data/flight-time.json")
19
20     flight_time_df = flight_time_raw_df \
21         .withColumn("FL_DATE", to_date("FL_DATE", "M/d/y")) \
22         .withColumn("CANCELLED", when(col("CANCELLED") == "1", True).otherwise(False))
23
24     flight_time_df.show(10)
25
26     input("Press Enter to Finish...")
```

The code defines a SparkSession, reads a JSON file into a DataFrame, applies a schema, and shows the first 10 rows. It also includes a prompt for the user to press Enter to finish.

You can run this application, and it will work. I added a Python input method, so my application is not terminating, and it is waiting for me. You can see in the output, there is a message “press enter to finish” don’t press enter yet. I am holding my application from termination to see the Spark UI.



Spark UI is available to your localhost:4040 port. So let me start the browser and visit localhost:4040. You can see the same Spark UI on your local machine. This UI is only available if your Spark application is running. As soon as your application finishes, the Spark UI goes away.





Thank You
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